

Solution Requirements

Date	29 June 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	4 Marks

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user authentication is required. The application is publicly accessible for flood prediction.	Low
2	Authorization levels	All users have the same access level to use the flood prediction system.	Low
3	External interfaces	The system interacts with a trained machine learning model and a Flask web interface. No external APIs are required.	Medium
4	Transactions processing	The system accepts weather parameters, processes them using the trained model, and returns the flood prediction result.	High
5	Reporting	The system displays flood prediction results (Flood / No Flood) through the web interface.	Medium

6	Business rules, etc.	The prediction is generated only after validating all required weather parameters. The model processes standardized input values before prediction.	High
7	Compliance to laws or regulations	The system follows general data privacy practices and does not collect or store personal user information.	Medium
8	<i>Other</i>	The system provides a simple and user-friendly interface for entering weather parameters and viewing prediction results.	Medium

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system should generate predictions quickly after user input.</i>	Prediction displayed in less than 2 seconds
2	Scalability	The application should support multiple prediction requests without affecting performance.	Supports multiple concurrent users

3	Security & Data Privacy	No personal user information is stored. Input data is processed only for prediction.	User data is not permanently stored
4	Reliability & Availability	The application should provide consistent prediction results whenever the server is running.	99% application availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on different devices.	Users can complete a prediction in less than 1 minute
6	Maintainability	The code should be modular and easy to update or improve with new machine learning models.	Easy integration of future model updates